

Midterm test material

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CREATE TYPE PERSON AS OBJECT(
name      varchar2(20),
surname   varchar2(30),
phone     number);

DROP TYPE PERSON;
DROP TABLE PEOPLE;

CREATE TABLE PEOPLE OF PERSON
(phone PRIMARY KEY);

INSERT INTO PEOPLE VALUES
(PERSON('Jane', 'Smith', 2999999));

INSERT INTO PEOPLE VALUES ('Cris', 'Robinson', 2999990);

SELECT VALUE(A)
FROM     PEOPLE A
WHERE    A.name='Jane';

SELECT * FROM PEOPLE;

UPDATE PEOPLE A
SET A = PERSON('Andrew', 'Smith', '3333333')
WHERE A.surname='Smith' and A.name='Jane';

CREATE TYPE PERSON AS OBJECT(
name      varchar2(20),
surname   varchar2(30),
phone     number);

CREATE TABLE EMPLOYEES(
Id          number(4) Primary key,
employee    PERSON,
position    varchar2(30),
salary      number(6,2));

INSERT INTO Employees VALUES (101, Person('John', 'Smith', '1111111'), 'Bookkeeper', 180.00);
INSERT INTO Employees VALUES (102, Person('Jane', 'Smith', '2222222'), 'CEO', 200.00);

SELECT * FROM Employees;
SELECT A.EMPLOYEE FROM EMPLOYEES A;
SELECT A.EMPLOYEE.NAME FROM Employees A;

UPDATE Employees A
SET A.employee.surname='Robinson'
WHERE id=101;

CREATE TYPE GRADE_TYPE AS VARRAY(3) OF number(2);

CREATE TABLE STUDENTS
(ID number,
GRADES GRADE_TYPE);

INSERT INTO STUDENTS VALUES( 1, GRADE_TYPE(6,7,8));

SELECT * FROM STUDENTS;

SELECT C.* FROM STUDENTS A, TABLE(A.GRADES) C WHERE A.ID = 1;
CREATE TYPE Object_type_Book AS OBJECT (
Title      VARCHAR2(50),
Author     VARCHAR2(30),
Num        NUMBER(6));

CREATE TYPE Table_Books AS TABLE OF Object_Type_Book;

CREATE TABLE Table_Educational_Books (
Speciality  VARCHAR2(20),
Student_Course  NUMBER(3),
Literature_List  Table_Books)
NESTED TABLE Literature_List STORE AS Nested_Table_Books;

INSERT INTO Table_Educational_Books VALUES ('Databases', 102,
Table_Books(
Object_Type_Book('Databases', 'J.Koks', 123),
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Object_Type_Book('DB technologies ', 'I.Ozols', 126)
)
);

SELECT Literature_List
FROM Table_Educational_Books;

Select b.Speciality, p.*
From Table_Educational_Books b, Table(b.Literature_List) p;

CREATE TABLE Employees2 (
    E_ID          number (5),
    SURNAME       varchar2 (20),
    SALARY        number (9, 2),
    POSITION       varchar2 (20) );

INSERT INTO EMPLOYEES2 VALUES(1, 'Koks', 200.50,'carpenter');
INSERT INTO EMPLOYEES2 VALUES(2, 'Zars', 180.00, 'secretary');

CREATE TYPE ADDRESS_T AS OBJECT(
    STREET   VARCHAR2(20),
    A_NUMBER NUMBER,
    CITY     VARCHAR2(20));

CREATE TYPE PERSON_T AS OBJECT (
    P_ID      NUMBER,
    SURNAME   VARCHAR2(15),
    P_NAME    VARCHAR2(15),
    ADDRESS_REF ADDRESS_T );

CREATE TABLE ADDRESSES_TABLE OF ADDRESS_T;

CREATE TABLE PEOPLE OF PERSON_T
(ADDRESS with ROWID scope IS ADDRESSES_TABLE);

INSERT INTO ADDRESSES_TABLE VALUES(ADDRESS_T('Stabu', 70, 'Riga'));

INSERT INTO PEOPLE VALUES(1, 'Koks', 'Juris', NULL);
SELECT Deref(A.ADDRESS) FROM PEOPLE A;

CREATE TYPE Student UNDER Person_T (
    student_ID      varchar2(10),
    course          number,
    s_group         varchar2(5));

CREATE OR REPLACE TYPE Employee_Type AS OBJECT(
    E_ID          number (5),
    SURNAME       varchar2 (20),
    SALARY        number (9, 2),
    POSITION       varchar2 (20) );

SELECT *
FROM salaries s JOIN bus_drivers d
ON s.driver_code = d.personal_code
MODEL
PARTITION BY (month)
DIMENSION BY (year, driver_last_name)
MEASURES (paid_salary)
RULES(
    paid_salary[2023,'Adkins'] = paid_salary[2023,'Adkins']*1.1
)
ORDER BY year;

--Various rule examples
SELECT *
FROM salaries s JOIN bus_drivers d
ON s.driver_code = d.personal_code
MODEL
DIMENSION BY (year, driver_last_name, month)
MEASURES (paid_salary)
RULES(
    paid_salary[2023,'Adkins','JAN'] = paid_salary[2023,'Adkins','JAN']*1.1,
    paid_salary[year > 2021,'Knowles','DEC'] = paid_salary[2023,'Knowles','JAN']*1.1,
    paid_salary[2023,'Bowie','JAN'] = MAX(paid_salary)[2023,'Adkins', month BETWEEN 'NOV' AND 'DEC'],
    paid_salary[2022, 'Mercury', month IN ('NOV', 'DEC')] = paid_salary[2022, 'Freidenfelds',
CV(month)]*1.5,
    paid_salary[2022, driver_last_name IS ANY, 'OCT'] = 10000
)
ORDER BY year;

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